

BUGFIXES

Bugfix Log

Revision: 12 **Last modified:** 2026-06-13T13:50:00Z

Per CONST-MD-Bugfix-Documentation, every bug surfaced during implementation gets a permanent entry below: title, root cause, affected files, fix description, regression guard.

Entries are append-only; do not edit historical entries except to add clarification footnotes.

2026-04-27 — boba-jackett implementation (plan 2026-04-27-jackett-management-ui-and-system-db.md)

1. Master-key two-step write silent data-loss window

Severity: HIGH (silent credential loss possible on crash mid-bootstrap).

Root cause: Initial implementation of `bootstrap.EnsureMasterKey` generated `BOBA_MASTER_KEY` in memory, returned it to the caller, and *then* wrote `.env` in a separate step. If the process crashed, was killed, or the host lost power between those two operations, the caller would proceed to encrypt credentials with a key that disk had no record of — making them permanently undecipherable.

Affected files: - `qBittorrent-go/internal/bootstrap/bootstrap.go` - `qBittorrent-go/internal/envfile/write.go`

Fix: Collapse the `generate + write` into a single `envfile.Atomic write (tmpfile + fsync + rename)` before the function returns. Caller never sees a key that hasn't been durably persisted. Commit `c0f20bc`.

Regression guard: - `qBittorrent-go/internal/bootstrap/bootstrap_test.go::TestEnsureMasterKeyDoesNotDuplicateHeader` - `qBittorrent-go/internal/envfile/write_test.go` atomic-write test layer

2. SQLite database file mode 0644 (world-readable credentials)

Severity: CRITICAL (any local user could read encrypted credentials + ciphertext-only attack surface).

Root cause: modernc.org/sqlite creates database files with the process's default umask, which on the container yielded 0644 (world-readable). The Go service's .env was already 0600 via envfile.Atomic, but boba.db was leaking — a defense-in-depth breach since the rows are encrypted, but still a violation of the principle “the DB file should be readable only by the service owner”.

Affected files: - qBittorrent-go/internal/db/conn.go

Fix: After the first Open() succeeds, db.Open calls os.Chmod(path, 0o600) (and the -wal / -shm siblings if present), collapsing the file mode to owner-only. Commit 8405167.

Regression guard: - Layer 4 security challenge challenges/scripts/boba_db_file_perms_challenge.sh - qBittorrent-go/internal/db/credential_leak_test.go

3. AutoconfigResult nil slices marshalled as JSON null

Severity: MEDIUM (silent client-breaking schema drift; UI rendered “undefined” badges instead of “0 items”).

Root cause: Go's json.Marshal serializes a nil slice as null, not []. When Autoconfigure() ran on a fresh DB with no discovered credentials, the discovered_credentials, configured_now, and already_present fields all marshalled as null. The OpenAPI 3.1 schema declared them as array, so clients (including the Angular dashboard) treated null as a contract violation.

Affected files: - qBittorrent-go/internal/jackett/autoconfig.go

Fix: Pre-allocate empty slices ([]string{}) at the top of Autoconfigure() so a “nothing to do” run still serializes as {..., "discovered": [], "configured_now": [], ...}. Commit 6f3dbaf.

Regression guard: - Layer 6 contract test qBittorrent-go/tests/contract/openapi_test.go (validates each named field is array per OpenAPI schema even on empty runs).

4. Catalog ReplaceAll empty-input wipe risk

Severity: HIGH (would permanently wipe the indexer catalog if Jackett returned an empty response; UI would show no indexers and fuzzy matcher would have nothing to match against).

Root cause: `repos.IndexerCatalog.ReplaceAll(rows []Row)` unconditionally DELETED existing rows then INSERTed the new ones. If rows was empty (Jackett momentarily unhealthy, transient 502, or catalog parse failure), the delete still happened, leaving an empty catalog table. Next refresh would have nothing to compare against.

Affected files: - `qBittorrent-go/internal/db/catalog.go` (or equivalent repo file)

Fix: Refuse the operation early with `errors.New("repos: ReplaceAll refusing empty replacement")`. The caller (autoconfig orchestrator) treats this as a soft error, logs it, and leaves the previous catalog intact. Commit 8d71df1.

Regression guard: - `qBittorrent-go/internal/jackettapi/catalog_test.go::TestRefreshCatalogAllTemplatesFailReturns200WithErrors`

5. `TestSearchHandler_QueueFull` pre-existing flake (NOT a boba-jackett bug)

Severity: LOW (test-only; flagged for follow-up).

Status: NOT FIXED in this plan. Confirmed unrelated to boba-jackett work.

Location: `qBittorrent-go/internal/api/api_test.go`

Symptom: Occasional intermittent failure under load when the search queue fills before the test's wait returns.

Action: Logged here for traceability so future audits don't mis-attribute it. Open issue suggestion: stabilise by injecting a deterministic queue-full hook instead of racing against real goroutine scheduling.

2026-06-09 — Session 7 fixes (`env_loader`, `AsyncMock`, `gamestorrents`)

6. `env_loader` flaky test: `KEY2` leak across test ordering

Severity: LOW (test-only flake; not a production bug).

Root cause: `test_comment_lines_ignored` in `tests/unit/test_env_loader.py` asserted `os.environ.get("KEY2")` is `None` after calling `load_env_files()` with a config file where `KEY2=commented_out` was commented out. However, `load_env_files` has a “first wins” policy — if `KEY2` was already present in `os.environ` from a prior test's `load_env_files` call, the function would not override it. Under `pytest-randomly` ordering, the test that sets `KEY2` sometimes ran before the comment-line test, causing intermittent failure.

Affected files: - `tests/unit/test_env_loader.py`

Fix: Added explicit `os.environ.pop("KEY1", None)` and `os.environ.pop("KEY2", None)` at the START of the test (before calling `load_env_files`), ensuring a clean env state regardless of test execution order. The `finally` block was retained as a safety net.

Regression guard: - `tests/unit/test_env_loader.py::test_comment_lines_ignored` — ran 2147/2147 twice consecutively under random ordering with zero failures.

7. AsyncMock warning in search deep-coverage tests

Severity: LOW (test warning; not a production bug).

Root cause: `test_iporrents_login_no_cookies` in `tests/unit/merge_service/test_search_deep_coverage.py` used `AsyncMock()` for `login_resp` and `mock_session`. These objects are used as context managers (async with `session.post(...)` as `resp`), not as directly awaited coroutines. `AsyncMock.__aenter__` returns a coroutine, but the test's mock setup didn't properly wire `__aenter__`/`__aexit__`, causing "coroutine was never awaited" `RuntimeWarning`.

Affected files: - `tests/unit/merge_service/test_search_deep_coverage.py`

Fix: Changed `AsyncMock()` to `MagicMock()` with explicit `__aenter__` and `__aexit__` stubs, matching the actual usage pattern (context manager, not coroutine).

Regression guard: - `tests/unit/merge_service/test_search_deep_coverage.py` — 0 `AsyncMock` warnings from this test (was 3).

8. gamestorrents _parse_size B-substring bug (documented, not fixed)

Severity: MEDIUM (plugin returns 0 for all realistic file sizes).

Root cause: Same class of bug as BOB-013 (torrentkitty). The multipliers dict in `_parse_size` iterates in insertion order: `{"B": 1, "KB": 1024, ...}`. Since "B" is a substring of "KB", "MB", "GB", and "TB", the check if unit in `size_str` matches "B" first for all realistic sizes. Then `size_str.replace("B", "")` leaves a trailing unit character (e.g. "35.2 G") which `float()` cannot parse, returning 0.

Affected files: - `plugins/gamestorrents.py` (line 74-86)

Fix: NOT FIXED in this session. Documented via tests that assert actual behavior (tests suffixed `_b_substring_bug`). Fix requires reordering the dict keys to check longest units first (same approach as BOB-013).

Regression guard: - `tests/unit/test_plugin_gamestorrents.py::TestParseSize` — 8 tests documenting the bug across GB/MB/KB/TB/commas/uppercase.

2026-06-09 — Session 8 parallel plugin tests + gamestorrents fix

9. gamestorrents `_parse_size` B-substring bug (fixed)

Severity: MEDIUM (plugin returned 0 for all realistic file sizes).

Root cause: Same class of bug as BOB-013 (torrentkitty). The multipliers dict in `_parse_size` iterated in insertion order: `{"B": 1, "KB": 1024, ...}`. Since "B" is a substring of "KB", "MB", "GB", and "TB", the check if unit in `size_str` matched "B" first for all realistic sizes. Then `size_str.replace("B", "")` left a trailing unit character (e.g. "35.2 G") which `float()` could not parse, returning 0.

Affected files: - `plugins/gamestorrents.py` (line 74-86)

Fix: Reordered dict keys to check longest units first: `{"TB": ..., "GB": ..., "MB": ..., "KB": ..., "B": 1}`. Same approach as BOB-013.

Regression guard: - `tests/unit/test_plugin_gamestorrents.py::TestParseSize` — 8 tests all pass with correct byte values for GB/MB/KB/TB/B/comma/uppercase.

10. piratebay `import os` placed after use (documented, not fixed)

Severity: LOW (only affects non-magnet torrent file downloads, which are rare).

Root cause: `piratebay.py:175` has `import os` placed after `os.fdopen` on line 172. When downloading a non-magnet torrent file, `os.fdopen` is called before `os` is imported, causing `UnboundLocalError`.

Affected files: - `plugins/piratebay.py` (line 172-175)

Fix: NOT FIXED in this session. Documented via test `test_torrent_file_download_known_bug_os_import_order`. Fix requires moving `import os` to the top of the `download_torrent` method or module level.

Regression guard: - `tests/unit/test_plugin_piratebay.py::TestDownloadTorrent::test_torrent_file_download_known_bug_os_import_order`

11. `torlock.search()` does not catch exceptions from `retrieve_url()`

Severity: LOW (network errors crash the entire search loop instead of failing gracefully).

Root cause: `torlock.py search()` calls `retrieve_url()` without a try/except, so a network error propagates up and crashes the search loop.

Affected files: - `plugins/torlock.py`

Fix: NOT FIXED in this session. Documented via test `test_search_exception_propagates`. Fix requires wrapping `retrieve_url()` in try/except with graceful error handling.

Regression guard: - tests/unit/
`test_plugin_torlock.py::TestSearch::test_search_exception_propagates`

2026-06-09 — Session 8 wave 2: nyaa/kickass/anilibra/torrentgalaxy/yts

12. nyaa download_torrent missing import re (documented, not fixed)

Severity: MEDIUM (any nyaa.si URL passed to `download_torrent` raises `NameError`).

Root cause: `plugins/nyaa.py:170` calls `re.search()` but `import re` is absent from the module. The `search()` method works because `re` is imported transitively via other modules, but `download_torrent` fails with `NameError: name 're' is not defined`.

Affected files: - `plugins/nyaa.py` (line 170)

Fix: NOT FIXED in this session. Documented via tests that assert the `NameError`. Fix requires adding `import re` at the top of the file.

Regression guard: - tests/unit/
`test_plugin_nyaa.py::TestDownloadTorrent` — 6 tests including the `NameError` documentation.

13. kickass comma-separated size not matched by regex (documented, not fixed)

Severity: LOW (sizes like `1,234.5 MB` silently parse to 0).

Root cause: The `HTMLParser` regex `\d+\.\d+` doesn't match comma-separated numbers like `1,234.5 MB`. The comma is not handled in the regex or the replacement logic.

Affected files: - `plugins/kickass.py`

Fix: NOT FIXED in this session. Documented via test `test_comma_in_size_not_matched_by_regex`. Fix requires updating the regex to handle commas.

Regression guard: - tests/unit/
`test_plugin_kickass.py::TestHTMLParserFeed::test_comma_in_size_not_matched_by_regex`

14. kickass crash-prone patterns (fixed)

Severity: MEDIUM (empty responses from `retrieve_url` crash the plugin).

Root cause: Three methods in `kickass.py` called `retrieve_url()` without `try/except` or empty-response guards: `__retrieve_download_link()`, `download_torrent()`, and `search()`. When `retrieve_url` returned `None` or empty string, subsequent `re.search/re.sub` calls crashed with `TypeError`.

Affected files: - `plugins/kickass.py` (lines 61, 74, 94)

Fix: Added `try/except` + empty-response guards to all three methods. Each now handles empty/`None` responses gracefully (returns “`NotFound`”, prints fallback URL, or breaks the loop respectively).

Regression guard: - `tests/unit/test_plugin_kickass_guards.py` — 13 tests proving empty response, `None` response, `ConnectionError`, `TimeoutError`, and malformed HTML don't crash.

2026-06-09 — Waves 5-8: B-substring epidemic across 9 community plugins

15. Systemic `_parse_size` B-substring bug (9 instances found, all fixed)

Severity: MEDIUM (all 9 plugins returned 0 for KB/MB/GB/TB — silently broken size reporting).

Root cause: The `_parse_size` method in each plugin uses a `multipliers` dict iterated in insertion order. With keys ordered `{"B": 1, "KB": 1024, ...}`, the check “B” in `size_str` matches the “B” substring inside “KB”, “MB”, “GB”, and “TB”. Then `size_str.replace("B", "")` leaves a trailing character (e.g. “1.5 G”) which `float()` cannot parse, returning 0.

Affected plugins (9 total): | Plugin | Status | |———|———| | `gamestorrents.py` | Fixed (BOB-024) | | `megapeer.py` | Fixed (BOB-042) | | `one337x.py` | Fixed (BOB-043) | | `extratorrent.py` | Fixed (BOB-044) | | `torrentfunk.py` | Fixed (BOB-045) | | `therarbg.py` | Fixed (BOB-047) | | `pctorrent.py` | Fixed (BOB-052, by subagent) | | `bitru.py` | Fixed (BOB-054) | | `xbsub.py` | Fixed (BOB-057) | | `yihua.py` | Fixed (BOB-058) |

Fix: Reorder dict keys longest-unit-first: `{"TB": ..., "GB": ..., "MB": ..., "KB": ..., "B": 1}`. Same approach as BOB-013 (`torrentkitty`).

Global regression guard: 10 dedicated test files, each with explicit `_parse_size` assertions for all 5 units (B/KB/MB/GB/TB).

16. Systematic `import re` omissions (3 instances found, all fixed)

Root cause: Three plugins called `re.search()` or `re.compile()` without importing `re` at module level. The `search()` method worked via transitive imports from helper modules, but `download_torrent()` failed with `NameError`.

Affected plugins: | Plugin | Status | |———|———| | `nyaa.py` | Fixed (BOB-035) | | `audiobookbay.py` | Fixed (BOB-042) | | (`torlock search()` also lacks exception handling — documented BOB-029) |

Fix: Added `import re` at module level.

17. `bt4g` infinite loop (test-only, fixed)

Severity: LOW (test hang, not production).

Root cause: `search()` has a `while True` loop that breaks when `parser.noTorrents` is `True`. Tests that used `return_value=MATCHING_HTML` (not `side_effect`) caused every page to match, triggering infinite loop.

Fix: Changed to `side_effect=[MATCHING_HTML, EMPTY_HTML]` so page 2 triggers the `noTorrents` break condition.

2026-06-13 — BobaLink extension: flaky-perf-test hardening + WCAG contrast fixes

Batch verified together by `extension/ci-ext.sh` → **CI-EXT: PASS**, full suite **632 passed (632)**, `tsc --noEmit clean`, `npm run lint 0/0`, `chrome+firefox` builds loadable, §11.4.38 `asset-verify` pass, both store zips ≥ 10 KiB. Every fix below carries a permanent regression guard (§11.4.135) that was RED before the fix.

18. Junk-flood DoS security test — load-coupled absolute-wall-clock FAIL-bluff

Severity: MEDIUM (flaky test — §11.4.1 FAIL-bluff; blocked the green gate).

Root cause (FACT, not guess): `extension/tests/security/scanner-hostile-input.test.ts` asserted a tight absolute `expect(elapsed).toBeLessThan(5000)` on a 50k-anchor orchestrator scan. On a shared/oversubscribed host (load-avg 15) the *correct* scan measured **6608 ms**; in isolation it passes < 5000 ms (proven 5/5). Product behaviour was always correct (exactly 2 magnets detected, all junk excluded) — only the absolute timing tripped.

Affected files: `extension/tests/security/scanner-hostile-input.test.ts`.

Fix: replaced the tight 5000 ms budget with a generous **30 s hang-ceiling** (catches a true hang / quadratic explosion — a real $O(n^2)$ over 50k anchors takes minutes) and moved the rigorous machine-independent DoS guard to a new perf test. Correctness assertions retained.

Regression guard: new extension/tests/perf/orchestrator-scaling.perf.test.ts asserts a metamorphic sub-quadratic scaling ratio ($t(10 \cdot N)/t(N) < 30$; linear ≈ 10 , quadratic ≈ 100 — dimensionless, immune to host load) with a golden-good/golden-bad oracle self-validation (§11.4.107(10)): linearRatio=9.9 accepted, quadraticRatio=113 rejected.

19. crypto.perf p99 budget — contention-coupled FAIL-bluff

Severity: MEDIUM (flaky test).

Root cause (FACT): extension/tests/perf/crypto.perf.test.ts asserted the 80 ms budget on dist.p99. PBKDF2 $\times 2$ is CPU-bound; under concurrent-suite contention p99 spiked to **164 ms** while intrinsic cost stayed ~ 23 –41 ms (proven in isolation). The newly-added heavy parallel perf/stress suites added the contention that exposed it.

Affected files: extension/tests/perf/crypto.perf.test.ts.

Fix: assert the budget on dist.min (intrinsic, contention-robust — contention only adds time). A real $\geq 3.6\times$ algorithmic regression still inflates the min past 80 ms; correctness (EXACT plaintext recovery) is asserted every round.

Regression guard: the same test (now contention-robust); isolated min ~ 23 ms ≤ 80 ; full-suite CI-EXT: PASS.

20. parsers.perf scaling-ratio — contention-coupled FAIL-bluff

Severity: MEDIUM (flaky test).

Root cause (FACT): extension/tests/perf/parsers.perf.test.ts computed the per-link scaling ratio from the **median** time at each size. The small (1000-link, ~ 5 ms) median is noise-sensitive under contention; the ratio tripped **3.09** > 3 while the intrinsic ratio is ~ 1.0 (proven 1.34 / 1.00 in isolation).

Affected files: extension/tests/perf/parsers.perf.test.ts.

Fix: compute per-link cost from the **min** run at each size (intrinsic). A genuine $O(n^2)$ regression still shows ratio ≈ 5 in the min run, so the cap of 3 keeps full regression-catching power while never flaking.

Regression guard: the same test; isolated ratio $1.24 \leq 3$; full-suite CI-EXT: PASS.

21. Three real WCAG 2.1 AA colour-contrast defects (popup theme tokens)

Severity: MEDIUM (accessibility — WCAG SC 1.4.3, real product defect).

Root cause (computed, §11.4.6): three popup theme tokens fell below the 4.5:1 normal-text contrast floor. Found by the new extension/tests/a11y/focus-and-contrast.a11y.test.ts (computes contrast ratios from the REAL committed CSS; analyzer self-validated against WCAG reference extremes), kept RED until fixed.

Affected files: extension/src/popup/styles.css (each token in both the @media prefers-color-scheme block and the explicit [data-theme] class).

Fix (same hue family, minimal visual change, ≥4.6:1): --text-faint dark on #1a1a2e: #7a7a90 (4.07) → #838399 (4.61) --text-faint light on #f5f6fa: #8a8a9c (3.14) → #6c6c7e (4.76) --warning-text light on #fff6e0: #9a6a00 (4.40) → #946400 (4.78)

Regression guard: the 3 a11y tests flipped RED→GREEN on the fix (proving they catch the defect) and are permanent §11.4.135 guards.

2026-06-13 — BobaLink extension wave-11: options-page WCAG a11y fixes

Found by the new extension/tests/a11y/options-contrast-motion.a11y.test.ts (computes contrast ratios from the REAL committed src/options/styles.css; analyzer self-validated against WCAG reference extremes), kept RED until fixed. Batch verified by extension/ci-ext.sh → **CI-EXT: PASS** with the wave-11 coverage additions. All RED→GREEN proven; the §11.4.120 reconciliation rewrote the gates to assert the NEW mechanism (not fake-passed, not reverted).

22. Save-button text below WCAG AA over its gradient (SC 1.4.3)

Severity: MEDIUM (accessibility, real). **Root cause (computed):** .btn-primary has color:#fff over linear-gradient(--accent → --accent-2); white on the lighter --accent (#667eea) end is only **3.66:1** (14px normal-weight text needs ≥4.5). **Fix:** the global --accent is ALSO .nav-item.active text on a dark sidebar where darkening it would REDUCE contrast, so the button uses a **button-local** darker indigo start #5a6fce (4.57:1); the --accent-2 end is already 6.37:1. **Affected:** src/options/styles.css .btn-primary. **Guard:** the test now reads the button's ACTUAL gradient stops and asserts #fff ≥ 4.5 over every stop.

23 & 24. Field labels + nav-hover invisible in light theme — hard-coded literals (SC 1.4.3)

Severity: MEDIUM (accessibility, real — labels effectively unreadable in light mode). **Root cause (computed):** .field > label (#d0d0e0) and .nav-item:hover (#e0e0e0) were hard-coded **dark-theme** literals that never re-resolved per theme → **1.40:1** and **1.12:1** on the light backgrounds. **Fix:** tokenized to theme-aware tokens — label → var(--text) (10.6/13:1), nav-hover → var(--text-strong) (15.5/16.1:1) — legible in both themes. **Affected:** src/options/styles.css. **Guard:** the §11.4.120-reconciled test asserts the selectors now use a var(--token) (not a literal) AND the resolved token clears AA in BOTH themes.

25 & 26. No prefers-reduced-motion escape (SC 2.3.3 / 2.2.2)

Severity: MEDIUM (accessibility, real). **Root cause:** the options CSS ships `@keyframes fadeIn` (panel reveal on tab activation) + transitions, and the popup CSS ships transitions, but neither had a `@media (prefers-reduced-motion: reduce)` block — motion-sensitive users had no way to suppress it. **Fix:** added the standard reduced-motion block (zeroing animation/transition durations) to BOTH `src/options/styles.css` and `src/popup/styles.css`. **Guard:** the ally tests assert each stylesheet ships non-trivial motion AND a reduced-motion block (RED→GREEN on the fix).

2026-06-13 — BobaLink extension wave-12: scaling-ratio FAIL-bluff hardening

Batch verified by `extension/ci-ext.sh` → **CI-EXT: PASS**, full suite **799 passed (799)** (+38 over wave-11: `crypto-tamper` 17, `link-scanner-coverage` 10, `highlight-manager` 11 — all green, no product defects). The one fix below was a flaky test exposed by the heavier 799-test load.

27. infohash dedup scaling-ratio test — single-run + tiny-floor FAIL-bluff (§11.4.50)

Severity: MEDIUM (flaky test — §11.4.1 FAIL-bluff; blocked the green gate). **Root cause (FACT):** `tests/security/infohash-detection-hostile.test.ts`'s relative-scaling guard used a SINGLE-run `tLarge / Math.max(tSmall, 0.05)`; on a sub-ms baseline the 0.05 ms floor made the ratio noise-dominated (**53.6** observed under full-suite contention vs ~ 4 intrinsic). Worse, the ≤ 40 threshold was too loose to even catch the quadratic it guards ($4\times$ input → linear ≈ 4 , quadratic ≈ 16 — both < 40). Passed at 761 tests, tripped at 799. **Fix:** measure the MIN over 7 reps at each size (intrinsic, contention-robust — host stalls only ADD time) and tighten the threshold to **10** (sits BETWEEN linear ≈ 4 and quadratic ≈ 16 , so it now genuinely catches an $O(n^2)$ regression while never flaking). Verified stable 5/5 in isolation + full-suite **CI-EXT: PASS**. **Affected:** `extension/tests/security/infohash-detection-hostile.test.ts`. **Note:** the sibling stress test's median estimator was initially assessed robust — but it ALSO flaked one wave later under heavier load (see entry 28), confirming that median is not enough; min is.

2026-06-13 — BobaLink extension wave-13: tab-group scaling median→min + working-tree hygiene

Batch verified by `extension/ci-ext.sh` → **CI-EXT: PASS**, full suite **814 passed (814)** (+15 over wave-12: `popup-states` 5 — genuine partial-Send-All gap; plus two prior-session test files brought into git — `options-save-flow` 4 + `offline-queue-persistence` 6 — that were sitting UNTRACKED in the working tree). No product defects.

28. tab-group scaling-ratio test — median estimator spikes under load (§11.4.50)

Severity: MEDIUM (flaky test — §11.4.1 FAIL-bluff; blocked the green gate).

Root cause (FACT): tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts's scaling guard used the MEDIAN of 5 reps; under the heaviest concurrent load (814 tests) several reps landed on a busy core and dragged the median up, so the 10×-input ratio spiked past the 25 ceiling while the intrinsic ratio is ~5–8. (This refutes the wave-12 assessment that median was robust — entry 27's note.) **Fix:** switched the shared timing helper from median to **MIN** over reps (the minimum is the intrinsic-cost estimator — host stalls only ADD time, so the fastest run is closest to true cost; a genuine $O(n^2)$ regression still inflates every run including the min) and bumped reps (orchestrator 3→5, tab-group 5→9). Verified stable 3/3 in isolation (ratios 4.76/6.41/8.25 vs the 25 ceiling) + full-suite CI-EXT: **PASS**. **Affected:** extension/tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts. **Lesson (§11.4.118):** the robust estimator for CPU-bound scaling-ratio tests under concurrent load is MIN-of-reps, not median/mean — all six flaky-scaling fixes this session converge on it.

2026-06-13 — BobaLink extension wave-14: live-7187 harness hardening

Live-harness prep to de-risk the operator-gated live round-trip. The default suite is unchanged at **814 passed (814)** (CI-EXT: **PASS**); the live test + challenge are out-of-suite and operator-run.

29. live_detect_send challenge — §11.4.14 cleanup-race leaves an orphan synthetic torrent

Severity: MEDIUM (anti-bluff/hygiene defect in a live Challenge). **Root cause (FACT):** challenges/extension/live_detect_send_challenge.sh cleaned up the synthetic torrent it adds ONLY when its secondary qBittorrent torrents/info probe reported present. But the proxy adds the magnet itself; that separate probe can race/auth-fail and read absent while the magnet IS added — so a PASS could leave an orphan synthetic torrent in qBittorrent (a §11.4.14 quiescence violation). **Fix:** cleanup now keys off the **proxy's authoritative per-url verdict** (results[0].status == "added") OR the probe, re-logs-in for a fresh SID when needed, and warns honestly if delete-login fails (never reds the run at teardown). Also fixed a /bin/sh -n parse issue (apostrophe/metachar in added comments — §11.4.67). **Affected:** challenges/extension/live_detect_send_challenge.sh. **Verification:** clean SKIP (exit 77) when the backend is down (no fail-open), bash -n + sh -n PASS, mutation-verified (no-op/partial response stubs → FAIL); the cleanup trigger confirmed to fire on a simulated proxy-added + probe-absent response (the old code would not). The up-path runs when the operator brings the stack up.

2026-06-13 — Boba broader-project (merge-service / Go / plugins / frontend) verification + 2 fixes

A full-project verification pass (parallel subagents, all run against the REAL toolchains, anti-bluff captured evidence) confirmed the broader Boba backend is green: **Go qBitTorrent-go 14 packages PASS under -race, vet clean, no data races; plugins: 60 .py compile, 12 curated contract-valid, 901 plugin tests pass** (under .venv/py3.13); **Angular frontend: build GREEN + 342 unit tests pass; merge-service Python: 4149 unit tests pass, ruff clean** (under .venv/py3.13). Two genuine items found:

30. ci.sh / scripts/run-tests.sh false-red under a host python3 < 3.12

Severity: MEDIUM (the project's own manual CI gate false-reds — a §11.4.1 FAIL-bluff at the toolchain layer). **Root cause (FACT, independently found by two subagents):** both scripts invoke a **bare python3**, which resolves to the host default. pyproject.toml declares requires-python = ">=3.12" and the suite uses tomllib + PEP-604 X | Y union syntax that a python3 < 3.12 cannot even COLLECT. On this host python3 = **3.9.25**, so python3 -m pytest tests/unit/ aborts: **"Interrupted: 36 errors during collection"** (TypeError: unsupported operand), while the project .venv (py3.13) collects **4149** and passes all. **Fix:** both scripts now select a >=3.12 interpreter via a _select_python helper (prefers \$PYTHON → .venv/bin/python → python3.13 → python3.12 → a bare python3 only if it is ≥3.12), aborting with a clear message if none qualifies. Backward-compatible: a host already on python3 ≥ 3.12 selects the same interpreter. **Affected:** ci.sh (lines for py_compile + the unit/integration/e2e pytest steps), scripts/run-tests.sh (the 5 exec ... pytest modes). **Verification (captured):** bash -n clean both; the selector picks .venv py3.13; bare py3.9 → "36 errors during collection" vs selected interpreter → "4149 tests collected" on the same tests/unit/ dir.

31. Go backend — 2 uncovered critical paths closed (coverage, not a defect)

Type: Task (coverage). The qBitTorrent-go merge_search RunSearch goroutine-orchestration loop (every prior test passed a nil client) and the sse_broker concurrent pub/sub had zero real coverage. Added 4 anti-bluff tests (internal/service/coverage_test.go, real httptest qBittorrent server, no mocks) — RunSearch completes/accumulates + StartFails→failed + ContextCancel→abort, and an 8-publisher/16-subscriber SSEBroker churn under -race. **Anti-bluff proof:** a no-op return nil stub in RunSearch made all three RunSearch tests FAIL, then reverted (production clean, §11.4.84). Verified GREEN under -race (service package ok 5.933s, no data race). Production code unchanged.

2026-06-13 — Boba merge-service hermetic specialized suites: 5 test-quality fixes (product correct)

A verification pass over the merge-service's hermetic specialized suites (stress, chaos, property, contract, concurrency, memory, observability, security — run under `.venv/py3.13`, the suites BG-1's tests/unit pass did not cover) found the PRODUCT is correct + secure (bandit 0 HIGH; CORS hardened; no product defect) but surfaced 5 real TEST-quality defects across the anti-bluff classes. All test-side only — `download-proxy/src/` unchanged; each verified green under `.venv`.

32. Five merge-service test-quality defects (§11.4.1 / §11.4.3 / §11.4.50 / §11.4.120)

- **tests/stress/test_pipeline_stress.py** — §11.4.50 flaky absolute-wall-clock threshold. Gated on `max(latencies_ms) < 2000` (absolute); a lone GC/scheduler stall spiked one iteration to ~140 ms (~25× the 5.6 ms median) — an absolute bound is the wrong oracle. Fixed to a RELATIVE bound `max ≤ 200 × max(p50, 1.0 ms)` (an $O(n^2)$ blowup inflates the median too, so it still fires; a lone outlier no longer flakes). Verified: 3 passed.
 - **tests/contract/test_crossapp_theme_contract.py** — §11.4.1/§11.4.3 FAIL-instead-of-SKIP. The 3 `@requires_compose` tests' own docstring promises "skip cleanly when the proxy is unreachable", but `_fetch()` did a bare `urlopen` → `URLError` → FAIL when down. Fixed `_fetch()` to catch only `CONNECTION` errors → `pytest.skip` (with `./start.sh -p reason`); `HTTPError` still returns the real status and `AssertionError` is NOT swallowed (a real missing-bridge still FAILs when the proxy is up). Verified: 3 SKIP when down, suite 27 passed when collectable.
 - **tests/security/test_cors.py** — §11.4.120 stale gate asserting REMOVED insecure behavior. `TestCORSWildcardDefault` asserted `Access-Control-Allow-Origin == "*" as the default`, but the product was deliberately hardened to a secure-by-default `localhost/dev` allowlist (wildcard removed, CONTINUATION known-issue #5). The TEST lagged the security improvement. Reconciled to assert the NEW secure default — a known `localhost` origin reflected EXACTLY, an unknown LAN origin NOT reflected and never `*`. Anti-bluff: a wildcard-revert fails the `!= "*" assertions (RED-on-old/GREEN-on-new)`, not a tautology. Verified.
 - **tests/observability/test_metrics_exist.py** — §11.4.3 infra-FAIL instead of SKIP. A `@requires_compose` test made an unconditional `httpx.get(:7187/metrics)` → `ConnectError` FAIL when down. Added the existing `@merge_service_required` skip guard → SKIPS cleanly. Verified.
 - **tests/security/test_jackett_autoconfig_secrets.py** — toolchain-path FAIL. `subprocess.run(["bandit"])` raised `FileNotFoundError` (bandit is in `.venv/bin`, not on bare PATH). Added a `_resolve_bandit()` helper (prefers the running interpreter's bin dir, then `shutil.which`, SKIP if genuinely absent). Product clean: `.venv/bin/bandit` on `jackett_autoconfig.py` → 0 HIGH findings. Verified.
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2026-06-13 — Boba broken-doc-links + perf-test container-boot (the last test-type suites)

Ran the remaining test types (performance/benchmark/load/docs) under `.venv`. Benchmark: 9 infra-free pass (500×–4760× threshold headroom — not flaky, correctly left unchanged) + 5 by-design error-not-skip infra-down (intentional). Load: only a Locust `locustfile.py` (no pytest tests). Two real items fixed (BUGFIXES 33, 34).

33. docs/ test caught 8 broken internal markdown links + 1 test FAIL-bluff

Severity: LOW-MEDIUM (broken doc links + a §11.4.1 test FAIL-bluff). **Root cause + fix (FACT):** `tests/docs/test_no_broken_links.py` (resolves links relative to each doc's own dir, the markdown convention) flagged 8 docs. Seven were genuine broken links — `docs/*_Summary.md` linked to siblings with a spurious `docs/` prefix (`[Architecture Summary](docs/Architecture_Summary.md)` in a doc already inside `docs/` → resolves to `docs/docs/...`). Fixed by dropping the prefix (`[(docs/X) → ](X)`); `docs/Scripts_Summary.md`'s `[(scripts/pre_build_verification.sh) → ](..scripts/...)` (the script is at repo-root). The 8th — a research doc — was a **TEST FALL-BLUFF**: the link extractor matched `[${tab.title}](${tab.url})` inside a **JS code block** (not a doc link). Fixed the extractor to STRIP fenced + inline code before matching. Anti-bluff verified (§1.1): a real broken link in **prose** still FAILs (the strip only removes code, not real links — not a tautology). **Affected:** 7 docs/`*_Summary.md` (+regenerated `.html/.pdf`), `tests/docs/test_no_broken_links.py`. **Verified:** `tests/docs` → 129 passed / 0 failed / 2 honest skip (mkdocs absent).

34. Performance test BOOTED CONTAINERS + hung 120 s on infra-down (§11.4.3 / no-boot)

Severity: MEDIUM (a test that actively boots infra + hangs). **Root cause (FACT):** `tests/performance/test_concurrent_search.py::TestConcurrentSearch` consumed a `merge_service_live` fixture that, on ports-down, ran `podman/docker compose up -d` then `_wait_for_port(timeout=120)` — `pytest-timeout` killed the wait → 7 setup-ERRORs + 2 min wall-clock, and it was ACTIVELY booting containers. The suite's siblings (`stress/security/observability`) all use the fast `merge_service_required` probe-and-SKIP guard. **Fix:** applied that same `merge_service_required` guard (1 s socket+/health probe, never boots containers) → 7 honest infra-down SKIPs. Non-tautology: when the stack is up the class still runs every real assertion (status 200, latency ceilings, p50/p95) unchanged. (The benchmark suite's separate error-not-skip design is INTENTIONAL and was left untouched per §11.4.122/§11.4.6.) **Affected:** `tests/performance/test_concurrent_search.py`. **Verified:** 7 SKIP cleanly, no container boot, `download-proxy/src/` unchanged.